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Stat 5D

HW 10 #

Section 4.2

① Let $X \sim Bin(7, 0.3)$

(a) $P(X=1)$:-

$$= \binom{7}{1} (0.3)^1 (1-0.3)^{7-1}$$

$$= 0.2470629 = 0.247$$

(b) $P(X=2)$:-

$$= \binom{7}{2} (0.3)^2 (1-0.3)^{7-2}$$

$$= 0.3176523 = 0.3177$$

(c) $P(X<1)$:-

$$= P(X=0)$$

$$= \binom{7}{0} (0.3)^0 (1-0.3)^7$$

$$= 0.0824$$

(d) $P(X>4)$:-

$$= P(X=5) + P(X=6) + P(X=7)$$

$$= \binom{7}{5} (0.3)^5 (1-0.3)^2 + \binom{7}{6} (0.3)^6 (1-0.3)^1 + \binom{7}{7} (0.3)^7 (1-0.3)^0$$

$$= 0.0288$$

(2) $\mu_x = E(X) = np$

$= 7(0.3) = 2.1$

(3) $\sigma_x^2 = np(1-p) = 7(0.3)(1-0.3)$

$\boxed{\sigma_x^2 = 1.47}$

(3)

(a) $P(x=2)$ when $x \sim Bin(4, 0.6)$

$= \binom{4}{2} (0.6)^2 (1-0.6)^{4-2}$

$= 0.3456$

(b) $P(x \geq 2)$ when $x \sim Bin(8, 0.2)$

$1 - P(x \leq 2) = 1 - \binom{8}{2}(0.2)^2 (1-0.2)^{8-2}$

$- \binom{8}{1}(0.2)^1 (1-0.2)^{8-1}$

$- \binom{8}{0}(0.2)^0 (1-0.8)^8$

$= 0.2031$

(c) $P(x \leq 2)$ when $x \sim Bin(5, 0.4)$

$= \binom{5}{2}(0.4)^2 (1-0.4)^{5-2}$

$+ \binom{5}{1}(0.4)^1 (1-0.4)^{5-1}$

$+ \binom{5}{0}(0.4)^0 (1-0.4)^5$

$= 0.6826$

(d) $P(3 \leq x \leq 5)$ when $X \sim Bin(6, 0.7)$

$= \binom{6}{3}(0.7)^3(1-0.7)^{6-3}$
$+ \binom{6}{4}(0.7)^4(1-0.7)^{6-4}$
$+ \binom{6}{5}(0.7)^5(1-0.7)^{6-5}$

$= 0.8119$

7

(a) $X \sim Bin(20, .15)$

$P(x=5) = \binom{20}{5}(0.15)^5(0.85)^{15}$

$= 0.1028$

(b) $P(X \leq 4) = P(x=0) + P(x=1) + P(x=2) + P(x=3)$

$= \binom{20}{0}(0.15)^0(1-0.15)^{20} + \binom{20}{1}(0.15)^1(1-0.15)^{19}$
$+ \binom{20}{2}(0.15)^2(1-0.15)^{18} + \binom{20}{3}(0.15)^3(1-0.15)^{17}$

$= 0.6477$

(c) $P(X=0)$:-

$$= \binom{20}{0} (.15)^0 (1-.15)^{20}$$

$$= 0.0388$$

(d) $\mu_X = E(X) = np = 20(.15) = 3$

(e) $\sigma_X^2 = np(1-p) = 20(.15)(.85) = 1.597$

(f)

(a) $X \sim (5, .6)$

$$P(X \leq 3) = \binom{5}{3}(.6)^3(1-.6)^2 + \binom{5}{2}(.6)^2(1-.6)^3 + \binom{5}{1}(.6)^1(1-.6)^4 + \binom{5}{0}(.6)^0(1-.6)^5$$

$$= 0.6630$$

(b) $P(X=3) + P(X=2)$:-

$$= \binom{5}{2}(.6)^2(1-.6)^3 + \binom{5}{3}(.6)^3(1-.6)^2$$

$$= 0.5760$$

(12)

(a) $X \sim B(n, p_1)$

P[defective and can not be repaired]

$= 0.2 \times 0.4 = 0.08$

(b) $P(X = 2) \quad X \sim B(20, .08)$

$= \binom{20}{2} (.08)^2 (1 - 0.08)^{20-2}$

$= 0.271090605$

(c) $= 0.27109$

(13)

(a) ~~$X \sim MN(1, .9, .06, .04)$~~

P(used imm) + P(cut)

$= 90\% + 6\% = 0.9 + 0.06$

$= 0.96$

(15) $X \sim MN(10, .9, .06, .04)$

$P(X \leq 9)$

$= 1 - P(X = 10)$

$= 1 - \binom{10}{9} (.9)^9 (1-.9)^1$

$= 0.0582$

(16)

(a) $P(X > 1) = 1 - P(X=0) + P(X=1)$

$= 1 - \binom{10}{1} (.10)^1 (1-.10)^9$

$- \binom{10}{0} (.90)^0 (1-.10)^{10}$

$= 0.2639$

(c)

(b) $P(X>1)$ $\quad X \sim Bin(10, .2)$

$= 1 - P(X=1) + P(X=0)$

$= 1 - \binom{10}{1} (.2)^1 (1-.2)^9$

$- \binom{10}{0} (.2)^0 (1-.2)^{10}$

$= 0.2396$

(c) $P(X>1)$ $\quad X \sim Bin(10, .02)$

$= 1 - P(X=1) + P(X=0)$

$= 1 - \binom{10}{1} (.02)^1 (1-.02)^9$

$- \binom{10}{0} (.02)^0 (1-0.02)^{10}$

$1 - 0.8170728907 -$

$= 0.34968$